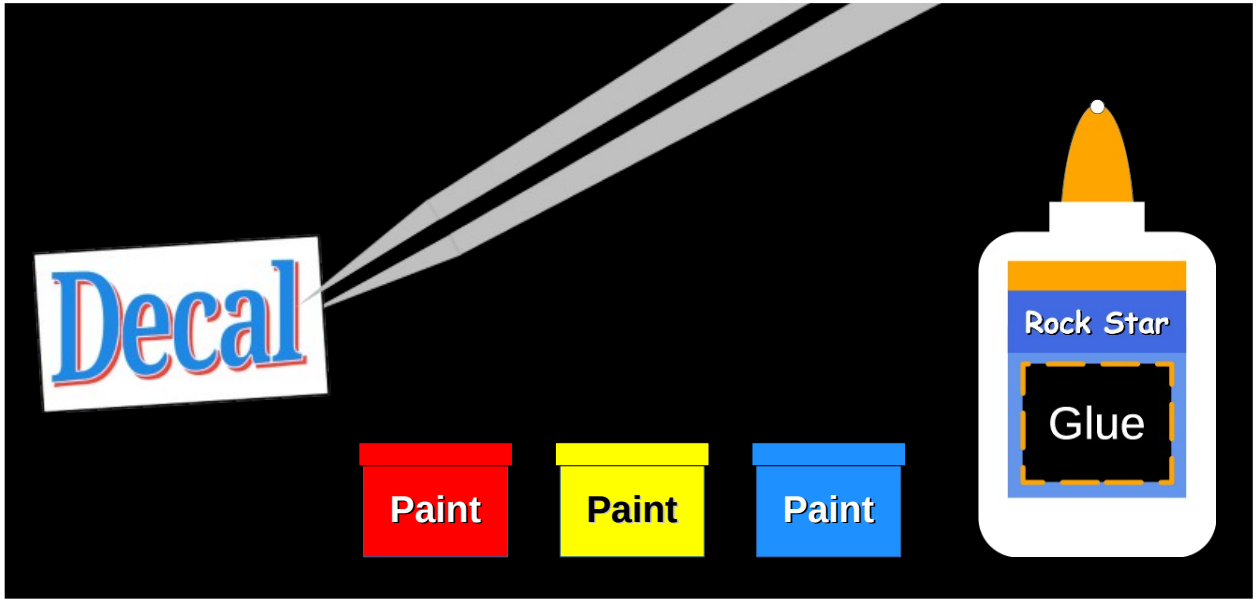
Cybersecurity Models



That Mostly Matter

About Me

I work for a big well-known organization...

As Vice President (VP) of Computer Security and Incident Response (IR). However, I have many years of hands-on technical experience, including Digital Forensics & Incident Response (DFIR).

https://www.linkedin.com/in/mattscheurer

https://x.com/c3rkah

Where I volunteer...

I am an Official

HACKING IS NOT A CRIME

Advocate
https://www.hackingisnotacrime.org

WARREN COUNTY CAREER CENTER

Advisory Board: Information Technology and Cybersecurity
https://www.mywccc.org

WOMSA
Empowering Women in Cybersecurity

Women's Security Alliance (WomSA) Technical Mentor
https://www.womsa.org

Disclaimer!

Yes, I have a day job. However…

Opinions expressed are based solely on my own independent security research and do not express or reflect the views or opinions of my employer.

Objective

The terms “Controls”, “Frameworks”, and “Models” are often used interchangeably.

While all are good, this talk is less about what I’d consider “Controls” or “Frameworks”; and focuses on what I consider Cybersecurity “Models”.

Agenda

- The classic “CIA triad”
- The “Cyber Kill Chain”
- The “Pyramid of Pain”
- Threat Modeling using “STRIDE”
- The “Diamond Model of Intrusion Analysis”

PDF

http://slides.dfirmatt.com

The Foundation

The ***CIA triad…***

CIA triad

CIA triad Missing Item?

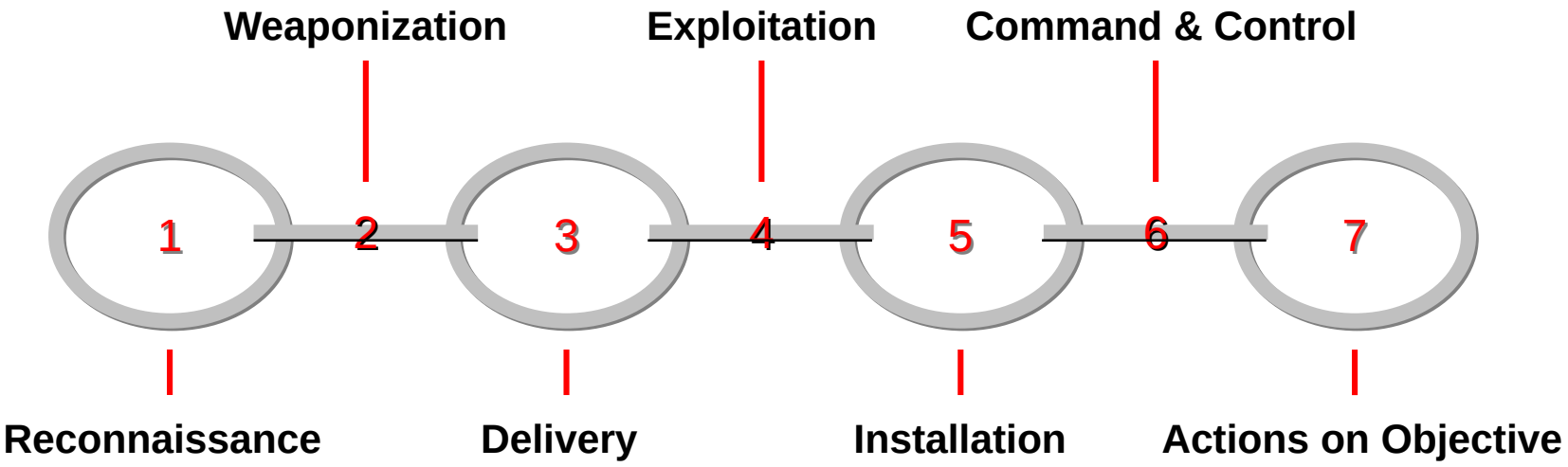
Non-Repudiation

The Stages of an Attack

*The **Cyber Kill Chain**…*

Source: Lockhead Martin

The Cyber Kill Chain

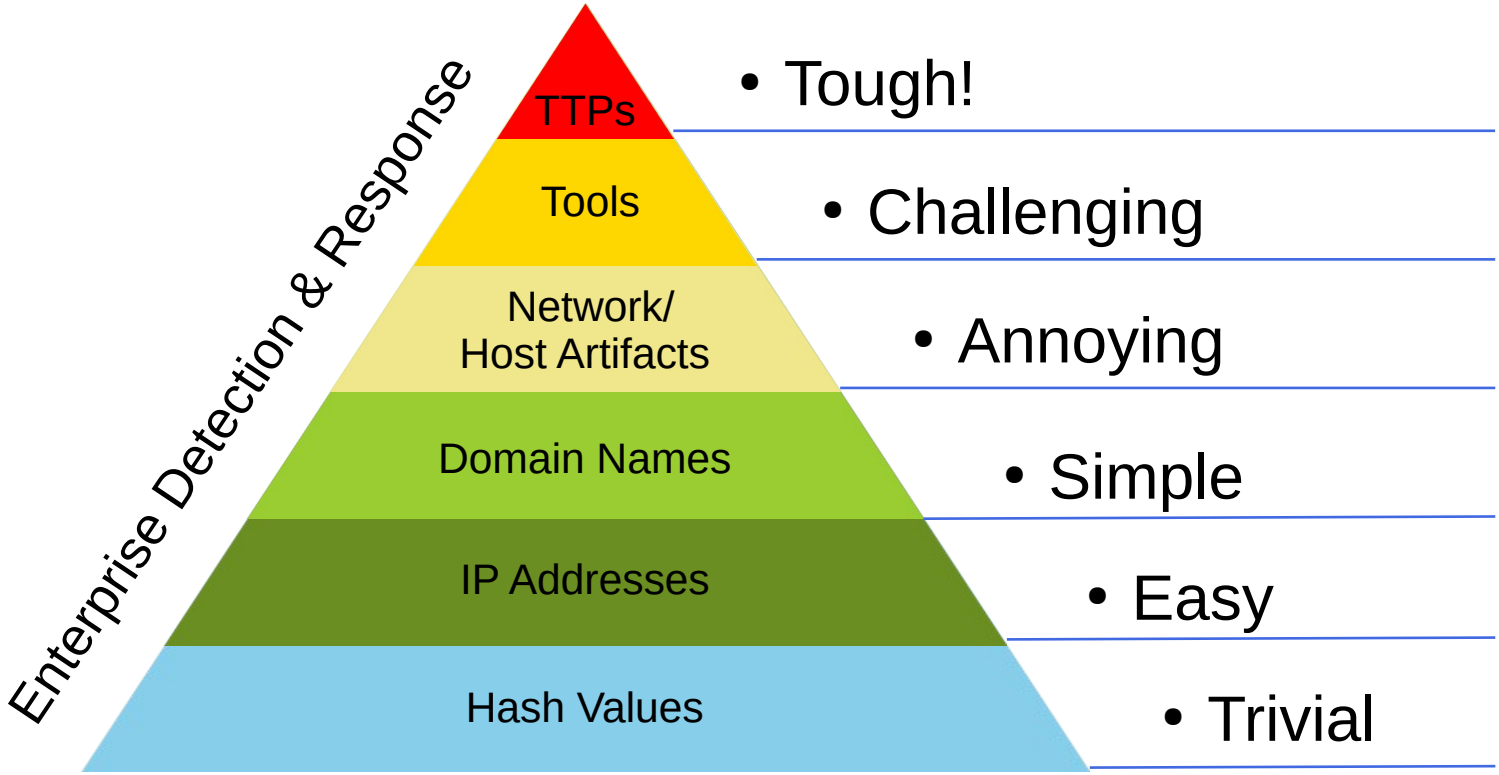


Detection Engineering

*The **Pyramid of Pain**…*

Source: David J. Bianco

The Pyramid of Pain



Threat Modeling

STRIDE *(or* ***STRIDE+****)*

One methodology, of many...

Source: Microsoft

The STRIDE Methodology

Spoofing
Tampering
Repudiation
Information Disclosure
Denial of Service
Elevation of Privilege

North Korea (DPRK) Examples

Reference Articles

https://www.justice.gov/opa/pr/justice-department-announces-coordinated-nationwide-actions-combat-north-korean-remote

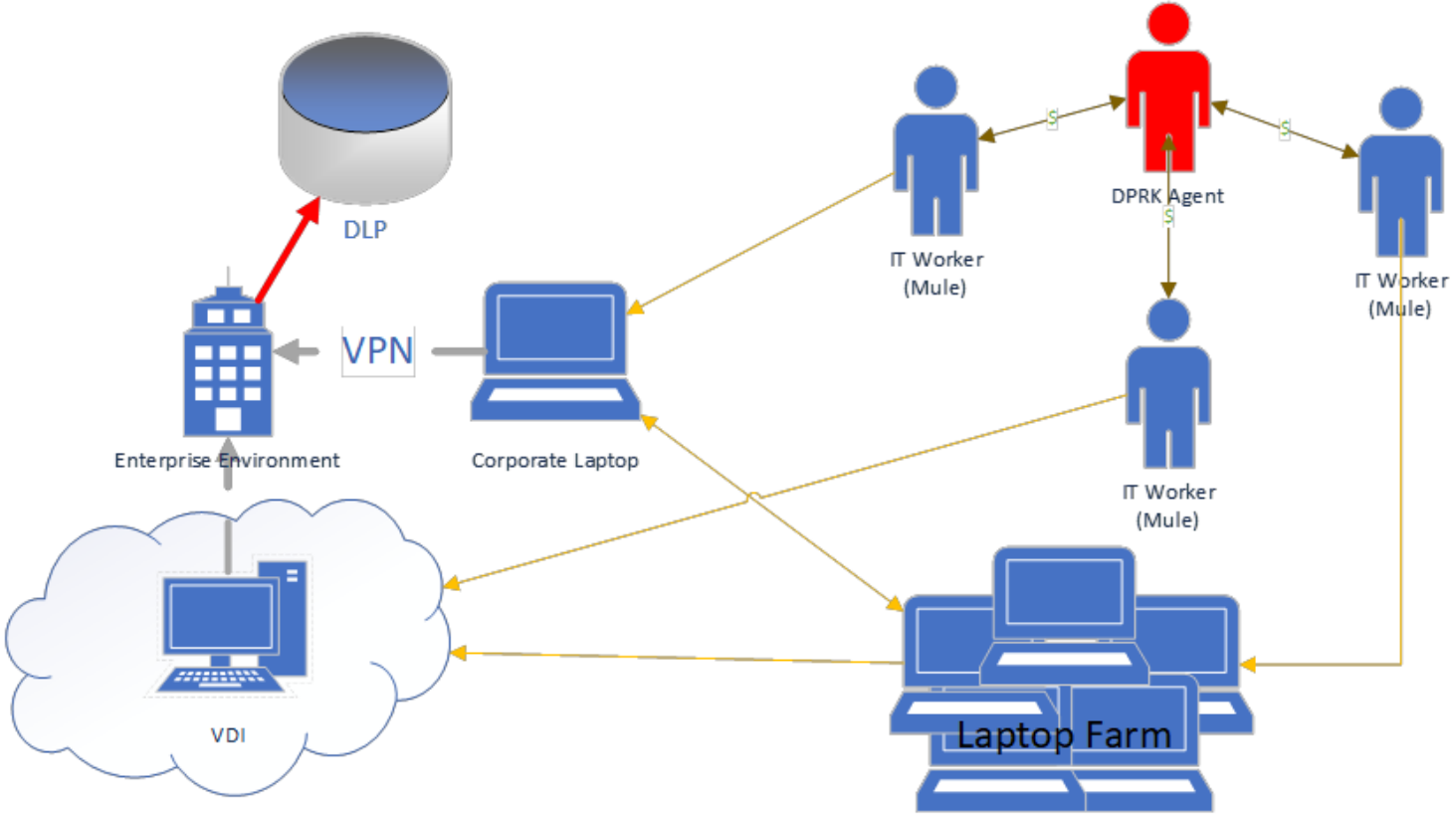
https://home.treasury.gov/news/press-releases/sb0190

Common Tradecraft

- Stolen Identities
- Laptop Farms
- RMM Applications
- VPNs
- Travel Routers
- KVM switches/PiKVMs

STRIDE “plus” Example

How do users, and also how does data get in and get out?



STRIDE to Risk Matrix Example

Threat	Risk Type	Likelihood	Importance	Risk Rating	Risk Score
Spoofing	Identity	Critical	High	High	10
Tampering	Device/Access	Critical	High	High	10
Repudiation	Evidence	Critical	Critical	Critical	12
Information Disclosure	Data Leakage	Critical	Critical	High	11
Denial of Service	Disruption	Low	Low	Low	3
Escalation of Privilege	Lateral Movement	Critical	High	Medium	9

Ratings: Critical (4), High (3), Medium (2) Low (1)

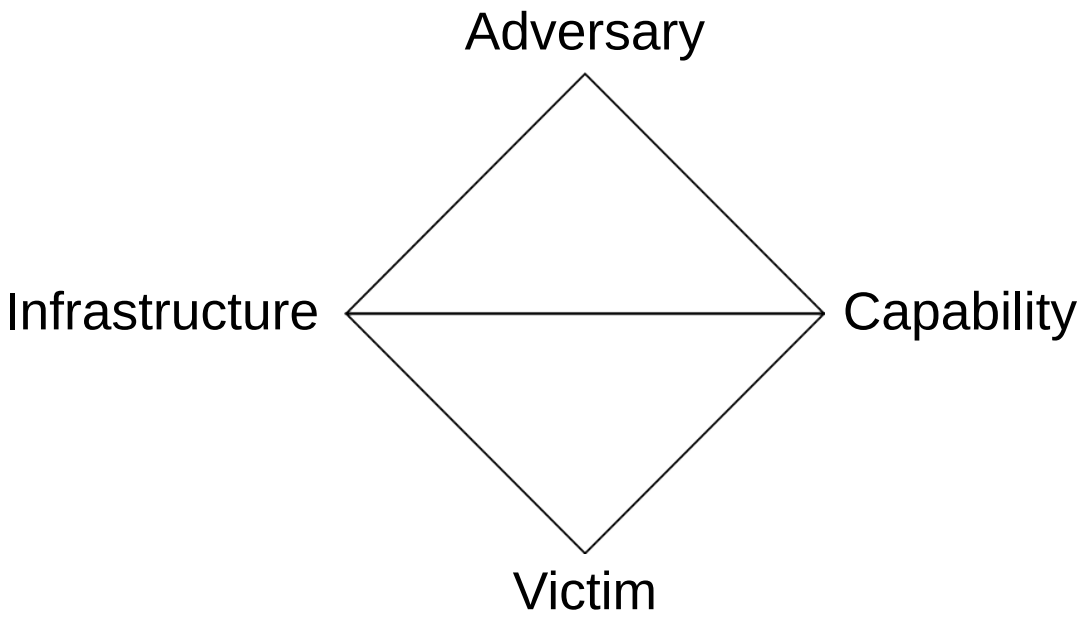
Intrusion Analysis

*The **Diamond Model**…*

Source: S. Caltagirone, A. Pendergast, & C. Betz

The Diamond Model

of Intrusion Analysis:



Diamond Model: Meta-Features

- Timestamp
- Phase
- Result

- Technology
- Socio-Political

- Direction
- Methodology
- Resources

Diamond Model Examples

https://thedfirreport.com/

Questions

Who?

What?

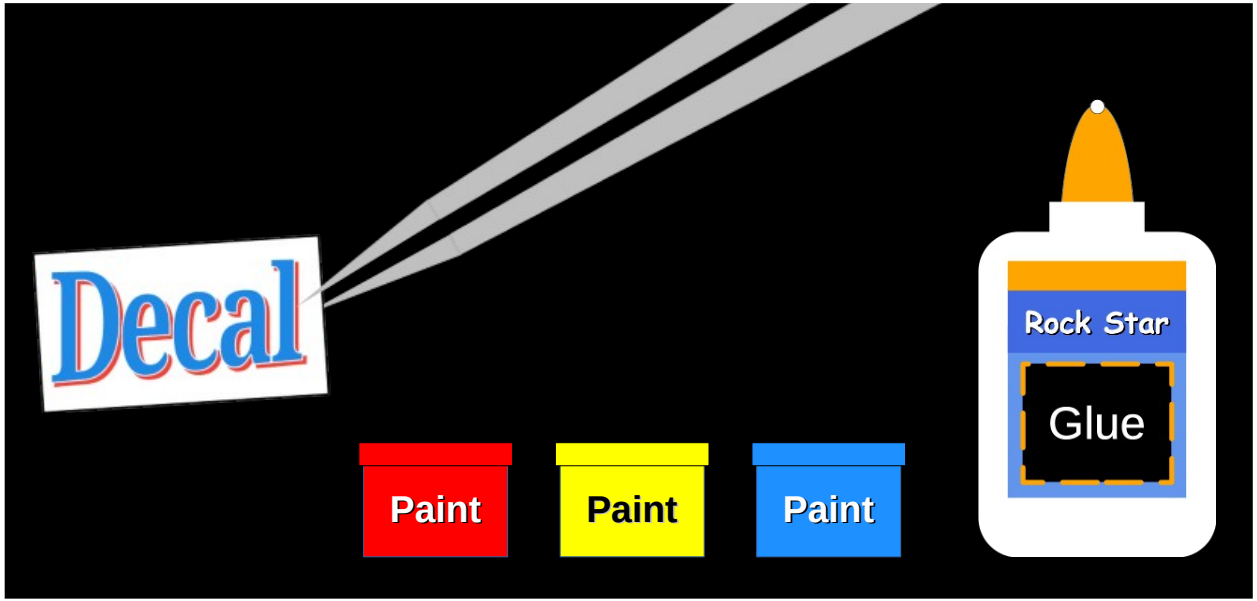
When?

Where?

Why?

How?

Cybersecurity Models



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